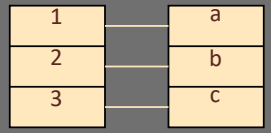
ERD Cardinality

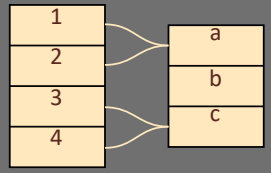
Maximum cardinality

One-to-one:



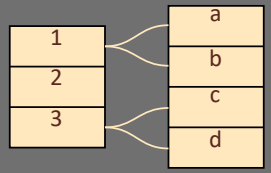


Many-to-one:



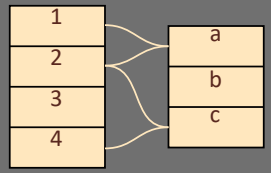


One-to-many:



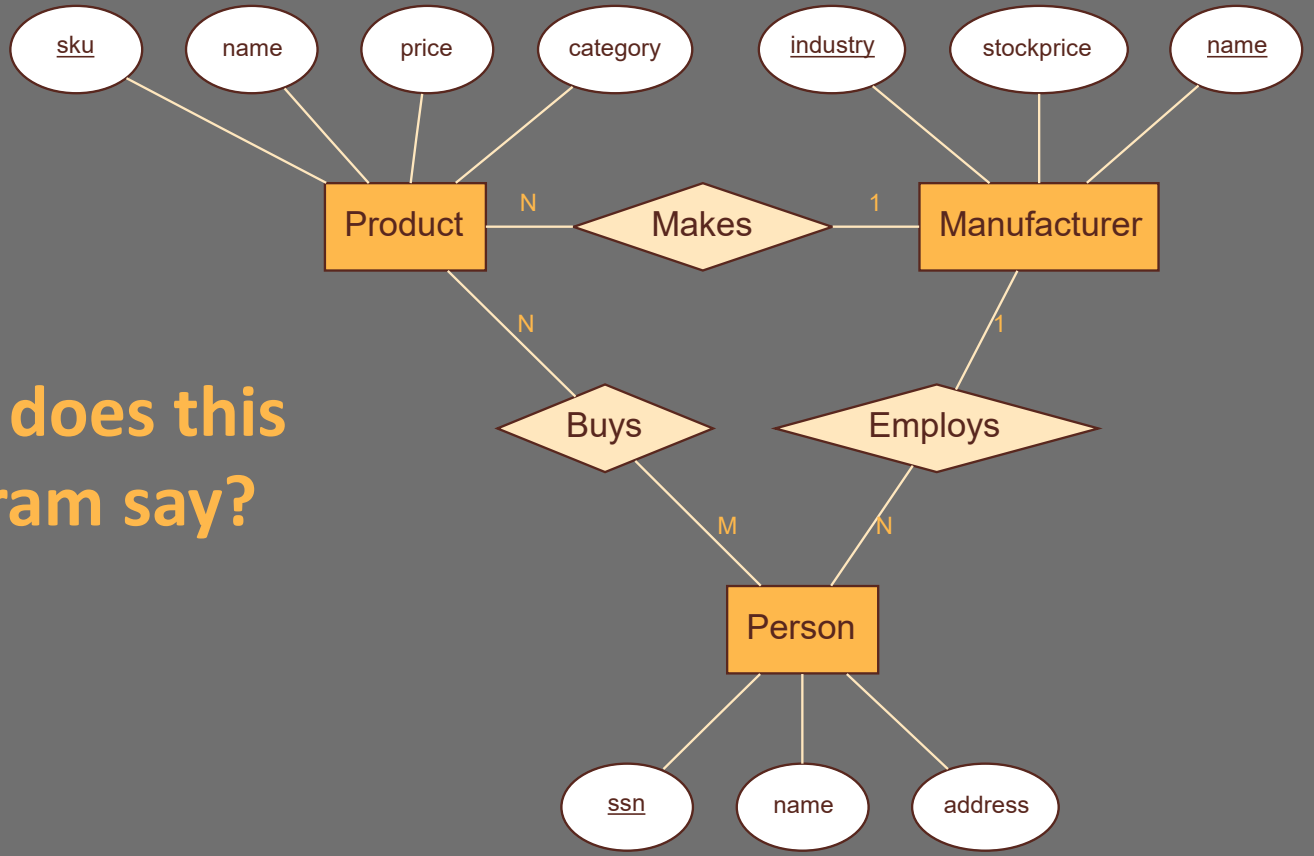


Many-to-many:





What does this diagram say?



Minimum cardinality (aka Participation)



Are there products made by no company?

Does every company make a product?



Or, simply N.

Each product maps to $1 \geq c \geq 1$ company.

Each company maps to $0 \geq p$ products.

Real-world cardinalities higher than expected

Expectation	Exceptions
Each person has at most one passport.	Dual citizenship
Each country has one capital.	Split executive & legislative capitals; Temporary capitals during war
Each country has one head of state.	Andorra, San Marino, …
Each President leads one country.	Simón Bolívar
Each student can take a course only once at a time.	Independent study courses

Other cardinalities allowed?

2:N, 1:5, …

No!